Miguel M.G. Ingeniería Informática

Gustave Doré - La visión del valle de los huesos secos, Ezequiel 37.1-2

FINAL: CÀLCUL/MATEMÀTIQUES

T: Miguel M. G.

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EXERCICI TEST

1. Sigui $P(t)$ la població d'una certa espècie a temps t. Si la seva dinàmica ve governada per l'equació.

$$\begin{cases} \frac{dP}{dt} = 50P(1 - \frac{P}{10000}) \\ P(t = 0) = 10 \end{cases}$$

$$\frac{1}{P - \frac{P^2}{10000}} dP = 50dt$$

$$10000 \int \frac{1}{10000P - P^2} dP = \int 50dt$$

$$10000 \int \frac{1}{10000P - P^2} dP = 10000 \int \frac{1}{P(10000 - P)} dP = 10000 \int \frac{A}{P} + \int \frac{B}{10000 - P}$$

$$1 = A(10000 - P) + BP$$

$P = 0$	$P = 10000$
$1 = 10000A$	$1 = 10000B$
$A = \frac{1}{10000}$	$B = \frac{1}{10000}$

$$10000 \cdot \frac{1}{10000} \left(\int \frac{1}{P} - \int \frac{-1}{10000 - P} \right) = ln \left| \frac{P}{10000 - P} \right| + C$$

$$ln \left| \frac{P}{10000 - P} \right| = 50t + C$$

$$\frac{P}{10000 - P} = Ce^{50t}$$

$$P(0) = 10$$

$$\frac{10}{10000 - 10} = C$$

$$\frac{1}{999} = C$$

$$\frac{P}{10000 - P} = \frac{e^{50t}}{999}$$

$$P = \frac{e^{50t}}{999} (10000 - P)$$

$$P\left(1 + \frac{e^{50t}}{999}\right) = \frac{10000e^{50t}}{999}$$

$$P = \frac{10000e^{50t}}{999\left(1 + \frac{e^{50t}}{999}\right)} = \frac{10000e^{50t}}{999 + e^{50t}} = \frac{10000}{999e^{-50t} + 1}$$

$$P(t) = \frac{10000}{999e^{-50t} + 1}$$

$$\lim_{t \to +\infty} \frac{10000}{999e^{-50t} + 1} = 10000$$

a) *té una única solució amb* $P(t) \leq 10000$

2. Sigui

$$F(x) := \int_0^{x^3} y^2 e^{y^5}\, dy$$

Regla de Leibniz

$$F(x) = \int_a^{g(x)} f(y)dy$$

$$F'(x) = f(g(x)) \cdot g'(x)$$

$$g(x) = x^3;\ g'(x) = 3x^2$$

$$f(y) = y^2 e^{y^5}$$

$$F'(x) = [(x^3)^2 e^{(x^3)^5}] \cdot 3x^2$$

d) $F'(x) = 3x^8 e^{x^{15}}$

3. L'àrea entre les corbes $y = -|x|$ i $y = 1 - 2x^2$ es pot calcular com

$$|x| = 2x^2 - 1$$

$$+ x$$

$$2x^2 - x - 1 = 0$$

$$\frac{1 \pm \sqrt{1-4\cdot 2\cdot(-1)}}{2\cdot 2} = \frac{1 \pm 3}{4};\ = 1;\ = \frac{-1}{2}$$

$$- x$$

$$2x^2 + x - 1 = 0$$

$$\frac{-1 \pm \sqrt{1-4\cdot 2\cdot(-1)}}{2\cdot 2} = \frac{-1 \pm 3}{4};\ = \frac{1}{2};\ = -1$$

$$\int_{-1}^{1} (1 - 2x^2) - (-|x|)dx = \int_{-1}^{1} 1 - 2x^2 + x dx = 2\int_0^1 -2x^2 + x + 1 dx$$

$$a)\ -\int_{-1}^{1} |x| - (1 - 2x^2)dx = \int_{-1}^{1} -2x^2 - x + 1 dx \rightarrow NO$$

$$b)\ \int_{-1}^{1} 2x^2 - 1 - |x| dx = \int_{-1}^{1} 2x^2 - x - 1 dx \rightarrow NO$$

$$c)\ \int_0^1 |x| - (2x^2 - 1)dx = \int_{-1}^{1} -2x^2 - x + 1 dx \rightarrow CORRECTE$$

$$d)\ \int_{-1}^{1} -|x| - (2x^2 - 1)dx = \int_{-1}^{1} -x - 2x^2 - 1 dx \rightarrow NO$$

$$e)\ 2\int_{-1}^{0} -|x| - (2x^2 - 1)dx = 2\int_{-1}^{0} -x - 2x^2 + 1 dx \rightarrow NO$$

Miguel M.G. Ingeniería Informática

4. La primitiva de la funció racional

$f(x) = \frac{1}{x^3-10x^2+16x-7}$

$F(x) = \int \frac{1}{x^3-10x^2+16x-7}dx$

Ruffini (no te preocupes es fácil ;))

	1	− 10	+ 16	− 7
+ 1		1	− 9	+ 7
	1	− 9	+ 7	0

$x^2 - 9x + 7 = 0$

$\frac{9\pm\sqrt{(-9)^2-4\cdot1\cdot7}}{2\cdot1} = \frac{9\pm\sqrt{53}}{2}$ $(no\ es\ un\ cuadrado\ perfecto)$

$F(x) = \int \frac{1}{(x-1)(x^2-9x+7)}dx = \int \frac{A}{x-1}dx + \int \frac{B}{x-\frac{9+\sqrt{53}}{2}}dx + \int \frac{C}{x-\frac{9-\sqrt{53}}{2}}dx$

$1 = A(x - \frac{9+\sqrt{53}}{2})(x - \frac{9-\sqrt{53}}{2}) + B(x - 1)(x - \frac{9-\sqrt{53}}{2}) + C(x - 1)(x - \frac{9+\sqrt{53}}{2})$

$x = 1$ $1 = A(\frac{-7-\sqrt{53}}{2})(\frac{-7+\sqrt{53}}{2})$ $\frac{4}{(-7)^2-(\sqrt{53})^2} = A$ $A = -1$	$x = \frac{9+\sqrt{53}}{2}$ $2 = B(\frac{7+\sqrt{53}}{2})(\sqrt{53})$ $B = \frac{2}{(7+\sqrt{53})(\sqrt{53})}$	$x = \frac{9-\sqrt{53}}{2}$ $-2 = C(\frac{7-\sqrt{53}}{2})(\sqrt{53})$ $C = \frac{-2}{(7-\sqrt{53})(\sqrt{53})}$

$\int \frac{A}{x-1}dx + \int \frac{B}{x-\frac{9+\sqrt{53}}{2}}dx + \int \frac{C}{x-\frac{9-\sqrt{53}}{2}}dx$

$-\int \frac{1}{x-1}dx + \frac{1}{(7+\sqrt{53})(\sqrt{53})}\int \frac{1}{x-\frac{9+\sqrt{53}}{2}}dx + \frac{-1}{(7-\sqrt{53})(\sqrt{53})}\int \frac{1}{x-\frac{9-\sqrt{53}}{2}}dx$

$-ln|x-1| + \frac{2}{(7+\sqrt{53})(\sqrt{53})}\cdot ln|x - \frac{9+\sqrt{53}}{2}| + \frac{-2}{(7-\sqrt{53})(\sqrt{53})}\cdot ln|x - \frac{9-\sqrt{53}}{2}| + C$

d) No te cap terme trigonomètric

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5. L'equació diferencial

$$\begin{cases} log(2)y''(x) + \sqrt{\pi}y'(x) + 2y = 0 \\ y(0) = 7,\ y'(0) = 10 \end{cases}$$

$y''(x) + \frac{\sqrt{\pi}}{log(2)}y'(x) + \frac{2}{log(2)}y = 0$

$p(x) = \frac{\sqrt{\pi}}{log(2)}\ i\ q(x) = \frac{2}{log(2)} \Rightarrow constantes$

Només una solució

Comprovació:

$log(2)x^2 + \sqrt{\pi}x + 2$

$\frac{-\sqrt{\pi}\pm\sqrt{(\sqrt{\pi})^2-4(log(2))(2)}}{2(log(2))}$

$\Delta < \pi - 8log(2)$

$r = \alpha \pm i\beta$

$\alpha = -\frac{\sqrt{\pi}}{2log(2)}$

$\beta = \frac{\sqrt{8log(2)-\pi}}{2log(2)}$

$y(x) = e^{\alpha x}(Acos(\beta x) + Bsin(\beta x))$

$y(0) = A = 7$

$y'(x) = e^{\alpha x}[(A\alpha + B\beta)cos(\beta x) + (B\alpha - A\beta)sin(\beta x)]$

$y'(0) = A\alpha + B\beta = 10$

$B = \frac{10-7\alpha}{\beta}$

$y(x) = e^{-\frac{\sqrt{\pi}}{2log(2)}x}(7cos(\frac{\sqrt{8log(2)-\pi}}{2log(2)}x) + \frac{10-7\alpha}{\beta}sin(\frac{\sqrt{8log(2)-\pi}}{2log(2)}x))$

b) té una única solució

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EXERCICI 2

a) Trobeu la solució general de l'edo:

$(sinx)(siny) + (cosy)y' = 0$

$(sinx)(siny) + (cosy)\frac{dy}{dx} = 0$

$cosy \cdot \frac{dy}{dx} = - sinx \cdot siny$

$\frac{cosy}{siny} dy = - sinx\, dx$

$\int \frac{cosy}{siny} dy = \int - sinx\, dx$

$ln|siny| = cosx + C$

$siny = Ce^{cosx}$

$y(x) = arcsin(Ce^{cosx})$

b) Trobeu la solució del problema de valors inicials següent

$y' + \frac{1}{x+100} y = 1$

$y(0) = 10$

$y'(t) + P(t)y = Q(t)$

$$y = e^{-\int P(x)dx} \left(\int Q(x) \cdot e^{\int P(x)dx} dx + C\right)$$

$\int P(x)dx = \int \frac{1}{x+100} dx = ln|x + 100| + C$

$y = e^{-ln|x+100|} (\int 1 \cdot e^{ln|x+100|} dx + C)$

$y = (\frac{1}{x+100})(\int x + 100 + C)$

$y = (\frac{1}{x+100})(\frac{x^2}{2} + 100x + C)$

$y(0) = 10$

$10 = \frac{1}{100} C$

$C = 1000$

$y = (\frac{1}{2(x+100)})(x^2 + 200x + 2000)$

$y = \frac{x^2+200x+2000}{2(x+100)}$

Miguel M.G. Ingeniería Informática

c) Trobeu la solució general de l'edo de segon ordre:

$y'' - 7y' + 12y = 4e^{2x}$

$x^2 - 7x + 12 = 0$

$x = \frac{7 \pm \sqrt{(-7)^2 - 4 \cdot 1 \cdot 12}}{2 \cdot 1} = \frac{7 \pm 1}{2}$

$x_1 = 4;\ x_2 = 3$

$(x - 3)(x - 4)$

$y_h(x) = Ae^{\alpha x} + Be^{\beta x}$

$y_h(x) = Ae^{3x} + Be^{4x}$

$y_p(x) = ke^{2x}$

$y_p'(x) = 2ke^{2x}$

$y_p''(x) = 4ke^{2x}$

$[4ke^{2x}] - 7[2ke^{2x}] + 12[ke^{2x}] = 4e^{2x}$

$2ke^{2x} = 4e^{2x}$

$k = 2$

$y = y_h + y_p = Ae^{3x} + Be^{4x} + 2e^{2x}$

EXERCICI 3

a) Trobeu una primitiva de la funció

$f(x) = \frac{x^2}{(x-1)(x+1)^2}$

$F(x) = \int \frac{x^2}{(x-1)(x+1)^2} dx = \int \frac{A}{x-1} dx + \int \frac{B}{x+1} dx + \int \frac{C}{(x+1)^2} dx$

$x^2 = A(x + 1)^2 + B(x - 1)(x + 1) + C(x - 1)$

$x = 1$ $1 = 4A$ $A = \frac{1}{4}$	$x = -1$ $1 = -2C$ $C = \frac{-1}{2}$	$x = 0$ $0 = \frac{1}{4} - B + \frac{1}{2}$ $B = \frac{3}{4}$

$\frac{1}{4}\int \frac{1}{x-1} dx + \frac{3}{4}\int \frac{1}{x+1} dx - \frac{1}{2}\int \frac{1}{(x+1)^2} dx$

$\frac{1}{4} ln|x - 1| + \frac{3}{4} ln|x + 1| + \frac{1}{2(x+1)} + C$

Miguel M.G. Ingeniería Informática

b) La figura limitada per la corba $y = xe^x$ i les rectes $y = 0$, $x = 1$ gira al voltant de l'eix OX.
Trobeu el volum del cos de revolució generat.
Recordeu que el volum de revolució generat per la gràfica de la funció f entre els punts $x = a$ i $x = b$ es pot calcular fent $\pi \int_a^b f(x)^2 dx$

$y = xe^x = 0$
$x = ln(0) = 0$
$x = 1$

$V_X = \pi \int_0^1 x^2 e^{2x}$

$\int udv = uv - \int vdu$

$u = x^2;\ du = 2x$
$dv = e^{2x};\ v = \frac{e^{2x}}{2}$

$\int x^2 e^{2x} = \frac{x^2 e^{2x}}{2} - \int 2x \frac{e^{2x}}{2}$

$u = 2x;\ du = 2$
$dv = \frac{e^{2x}}{2};\ v = \frac{e^{2x}}{4}$

$\int 2x \frac{e^{2x}}{2} = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4}$

$\int x^2 e^{2x} = \frac{x^2 e^{2x}}{2} - \frac{xe^{2x}}{2} + \frac{e^{2x}}{4}$

$V_X = \pi \int_0^1 x^2 e^{2x} dx = \pi[\frac{x^2 e^{2x}}{2} - \frac{xe^{2x}}{2} + \frac{e^{2x}}{4}]_0^1 = \pi[(\frac{e^2}{4}) - (\frac{1}{4})] = \pi \frac{e^2-1}{4} u^3$

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